

# Complete DevOps Workflow (Git, Terraform, Jenkins, Docker, Kubernetes, Datadog, AWS)

## 1. Git - Version Control System

- Git stores and manages application code.
- Developers push code to repositories like GitHub.
- Git tracks changes, supports team collaboration, and enables rollbacks.

## 2. Terraform - Infrastructure as Code (IaC)

- Terraform creates AWS infrastructure using .tf files.
- Commands: terraform init -> plan -> apply
- Example: Provision EC2, VPCs, RDS, etc.

## 3. Jenkins - CI/CD Pipeline

- Automates build, test, and deployment.
- Triggered by Git push via webhook.
- Stages: Build (Docker), Test, Deploy (Kubernetes/AWS)

## 4. Docker - Containerization

- Packages app and its dependencies into a container.
- Images are stored in DockerHub or Amazon ECR.
- These containers are deployed into Kubernetes pods.

## 5. Kubernetes - Container Orchestration

- Manages containers in clusters.
- Runs applications in Pods (smallest deployable units).
- Handles scaling, rolling updates, auto-restarts.

## 6. Pods in Kubernetes

- Created using deployment.yaml or via Jenkins.
- Use container images from registries.
- Kubernetes schedules them on nodes.

## 7. Datadog - Monitoring & Logging

- Installed as agents on EC2/Kubernetes.
- Monitors system metrics, logs, and performance.
- Logs are stored in Datadog Log Explorer for searching and alerting.

## 8. AWS - Cloud Infrastructure

- Terraform provisions infrastructure (EKS, EC2, RDS, etc.)
- Jenkins deploys Docker containers to Kubernetes on AWS.
- Datadog monitors applications running on AWS.

### Workflow Summary:

Developer -> Git Push -> Jenkins CI/CD -> Docker Build -> Kubernetes Deploy -> Terraform Infra -> Datadog Monitor -> AWS Hosts Everything

### Real-Life Analogy:

- Git = Recipe Book
- Terraform = Builder of the kitchen
- Jenkins = Automated Chef
- Docker = Meal box
- Kubernetes = Restaurant Manager
- Datadog = CCTV + Feedback system
- AWS = The building for the restaurant